

Alejandro Sánchez Yalí

AI Full Stack Engineer



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SUMMARY

AI Full Stack Engineer with 8+ years designing and shipping web applications end to end and 3+ years integrating LLMs into production products. Sole architect and engineer of three live systems – an AI WhatsApp support platform, an AI-assisted recruitment platform and a surgical supply chain system – spanning Python/FastAPI backends, Next.js/TypeScript frontends, PostgreSQL and cloud deployment. M.Sc. in Mathematics and 11 years teaching, which shows up in design docs, code review and mentoring.

TECHNICAL SKILLS

- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL
- **Frontend:** React.js (7+ years), Next.js (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, Effect, Three.js
- **Backend:** FastAPI, Django, Django REST Framework, Flask, NestJS, Node.js, Express, RESTful APIs, GraphQL, microservices
- **AI & LLMs:** LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, PyTorch, TensorFlow
- **Databases & Infra:** PostgreSQL, Redis, MongoDB, MySQL, Firebase, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Engineering:** Clean architecture, DDD, SOLID, TDD, Vitest, Jest, React Testing Library, code review

EXPERIENCE

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| AI Specialist
<i>Lapzo</i> | Aug 2025 – Present
<i>Remote, Mexico</i> |
| <ul style="list-style-type: none">– Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides– Owned the architecture connecting LLM services, ElevenLabs voice synthesis and the real-time layer built with Next.js, NestJS and Firebase Realtime Database– Lead member of the AI Committee, standardising AI development practices across the organisation | |
| Independent Software & AI Consultant
<i>Freelance</i> | Aug 2024 – Present
<i>Remote</i> |
| <ul style="list-style-type: none">– Delivered end-to-end web applications for business clients across both halves of the stack: React.js/Next.js/TypeScript on the front, Python/FastAPI/Django and Node.js/NestJS behind it– Lead engineer of Aluna (AI recruitment platform, see below) and builder of a multi-tenant AI customer service agent for CREARIA (Colombia) on WhatsApp Cloud API, LiteLLM, RAG and MCP with human escalation– Led technical design sessions with clients and made the technology decisions for each project; code reviewer for ROCKET CODE's development team– Deployed on AWS (EC2, S3, Docker) with CI/CD pipelines and Stripe payment integrations | |
| Full Stack Engineer
<i>Monadical</i> | Apr 2021 – Aug 2024
<i>Remote, Canada/US</i> |
| <ul style="list-style-type: none">– Built and maintained full stack applications in Python (Django, FastAPI) and TypeScript (React.js, Next.js), including RESTful and GraphQL APIs– Led frontend architecture, reusable component libraries and testing strategy (Vitest, Jest, TDD)– Integrated AI/ML services and LLMs (OpenAI, LangChain, Whisper) into production web apps– Led code reviews and mentored engineers on a distributed team across Canada, the US and Latin America | |
| Full Stack Engineer
<i>BCFort</i> | Aug 2018 – Oct 2020
<i>Medellín, Colombia</i> |
| <ul style="list-style-type: none">– Designed system architecture for blockchain platforms and built decentralised app frontends with React and Web3.js– Developed analytics platforms with Python (Django) and Node.js, managing cloud infrastructure on AWS, GCP and IBM Cloud | |

- Taught Machine Learning, Data Science and Mathematics with Python, NumPy, Pandas, TensorFlow and PyTorch
- Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research projects

FEATURED PROJECTS

- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, RAG, PostgreSQL, Redis, Next.js*
 - Sole architect and engineer of a multi-tenant SaaS where AI agents resolve routine customer inquiries in about 2 seconds and escalate the rest to human operators with full conversation context
 - Modular monolith with 10 DDD bounded contexts enforced by a lint rule in CI; multi-provider LLM layer (Groq default, OpenAI fallback) so switching providers is a config change
- **VitaStock – Surgical Supply Chain for Clinics**  – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
 - Sole architect and engineer of a platform for private surgical clinics in Colombia – 12 modules, 5 user roles, replacing spreadsheet-based inventory and procurement
 - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Aluna – AI Recruitment Platform for Staffing Agencies**  – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
 - Lead engineer of an AI recruitment platform for staffing agencies, built in 3 months – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
 - Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email

OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- 486 GitHub stars across own open-source projects – top 4.9% of ranked GitHub profiles (GitRanks)

TALKS & WRITING

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024
- **Building Plixiq: a Multi-Tenant WhatsApp AI Support Platform** – architecture case study (2026)

EDUCATION

M.Sc. in Mathematics
Universidad de Antioquia

Aug 2010 - Sep 2013
Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics
Universidad de Antioquia

Jan 2004 - Oct 2009
Medellín, Colombia